

SERVING GEOSPATIAL DATA USING MODERN AND LEGACY STANDARDS: A CASE STUDY FROM THE URBAN HEALTH DOMAIN

***** (for review, names must be rendered anonymous)

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Commission XX, WG XX/YY

KEY WORDS: Geospatial Data, Standards, Interoperability, API, Spatial Data Infrastructure, Urban Health, pygeoapi, Metadata

ABSTRACT:

The eMOTIONAL Cities project sets out to understand how the natural and built environment can shape the feelings and emotions of those who experience it. It does so with a cross-disciplinary approach which includes urban planners, doctors, psychologists, neuroscientists and engineers. At the core of this research project, lies a Spatial Data Infrastructure (SDI) which assembles datasets that characterise the emotional landscape and built environment, in different cities across Europe and the US. The SDI is a key tool, not only to make the research data available within the project consortium, but also to allow cross-fertilisation with other ongoing projects and later on, to reach a wider public audience. For more than twenty years SDIs have adopted the OGC W*s service interfaces, which are based on SOAP, the Simple Object Access Protocol. In recent years a new “family” of APIs has emerged within OGC, which is more aligned with modern web practices. In this project, we set out to leverage the advantages of this new approach, and compiled a stack to implement an SDI based on OGC APIs. However, we realised that we still need to support the legacy standards, either because an OGC API replacement is not mature enough, or there are no implementations available. This has led us to compile another stack based on the legacy standards. In this paper we describe our architecture, along with the challenges that we had to address. Both stacks are based on OSGeo Software, and they are available on GitHub.

1. MANUSCRIPT

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The paper should have the following structure:

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7. Conclusions
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Setting	A4 size page	
	mm	inches
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Column Width	82	3.2
Column Spacing	6	0.25

Table 1. Margin settings for A4 size page.

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The title should appear centred in bold capital letters, at the top of the first page of the paper with a size of twelve (12) points and single-spacing. Author(s) name(s), affiliation and mailing

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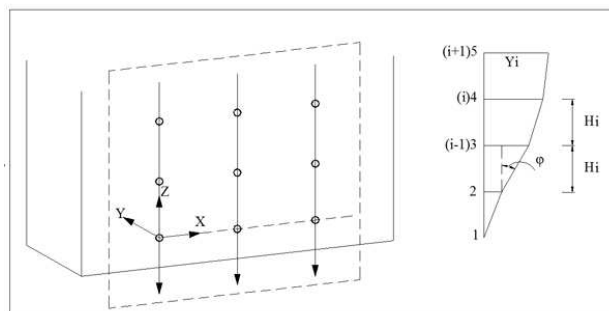


Figure 1. Figure placement and numbering.

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3.4.1 Equations: Equations should be numbered consecutively throughout the contribution. The equation number is enclosed in parentheses and placed flush right. Leave one blank lines before and after equations:

$$x = x_0 - c \frac{X - X_0}{Z - Z_0}; y = y_0 - c \frac{Y - Y_0}{Z - Z_0}, \quad (1)$$

where c = principle distance
 x, y = image coordinates
 X_0, Y_0, Z_0 = coordinates of projection centre
 X, Y, Z = object coordinates

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ACKNOWLEDGEMENTS (OPTIONAL)

Acknowledgements of support for the project/paper/author are welcome if they do not violate the anonymity of the submitted paper. Otherwise, they should be masked.

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